**Library Management System using Object-Oriented Programming Concepts in Java**

**Objective / Aim**

To understand and implement the core Object-Oriented Programming (OOP) concepts in Java, including Classes, Objects, Constructors, Encapsulation, Inheritance, Polymorphism, Abstraction, and Interfaces by developing a Library Management System.

**Algorithm / Logic Description**

**Part A – Classes and Objects**

1. Create a class named **Book**.
2. Declare Book ID, Book Name, Author, and Price.
3. Create an object and display its details.

**Part B – Constructors and Encapsulation**

1. Declare data members as **private**.
2. Create default and parameterized constructors.
3. Create Getter and Setter methods.
4. Access private data through methods.

**Part C – Inheritance**

1. Create a parent class **Person**.
2. Create child classes **Student** and **Faculty**.
3. Inherit the properties of Person.
4. Display inherited information.

**Part D – Polymorphism**

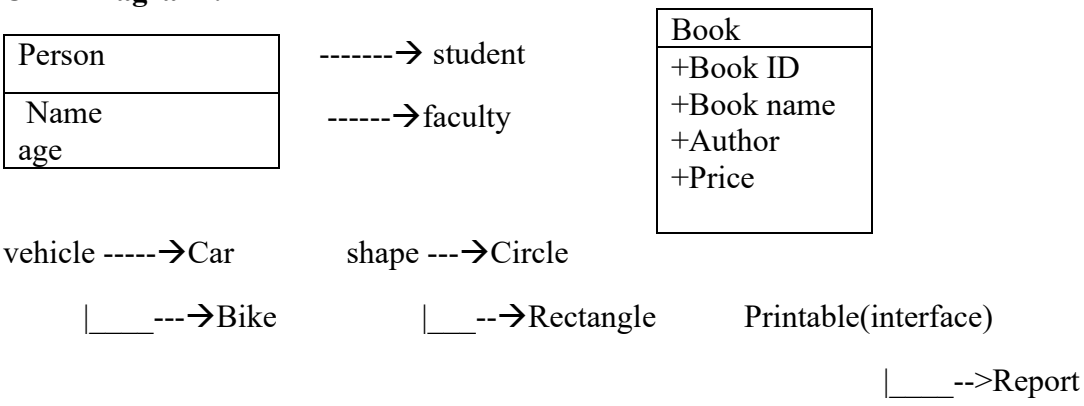
1. Implement method overloading using an **Area** class.
2. Implement method overriding using **Vehicle**, **Car**, and **Bike** classes.
3. Display overridden methods.

**Part E – Abstraction and Interfaces**

1. Create an abstract class **Shape**.
2. Implement draw() in **Circle** and **Rectangle**.
3. Create an interface **Printable**.

4. Implement the interface in **Report** class.
5. Display all outputs.

**UML Diagram :**




**Source code :**

```
package tasks;
public class LibraryManagementSystem {
    public static void main(String[] args) {
        System.out.println("------ BOOK DETAILS ------");
        Book b1 = new Book(101,"Java Programming","James Gosling",550);
        b1.display();
        System.out.println();
        System.out.println("------ STUDENT ------");
        Student s = new Student("Mohana",19,101);
        s.display();
        System.out.println();
        System.out.println("------ FACULTY ------");
        Faculty f = new Faculty("Ramesh",40,"Java");
        f.display();
        System.out.println();
```

```
        System.out.println("------ METHOD OVERLOADING ------");
        Area a = new Area();
        System.out.println("Circle Area = "+a.calculateArea(5.0));
        System.out.println("Rectangle Area = "+a.calculateArea(5,4));
        System.out.println("Triangle Area = "+a.calculateArea(10.0,8.0));
        System.out.println();
        System.out.println("------ METHOD OVERRIDING ------");
        Vehicle v1 = new Car();
        Vehicle v2 = new Bike();
        v1.display();
        v2.display();
        System.out.println();
        System.out.println("------ ABSTRACTION ------");
        Shape c = new Circle();
        Shape r = new Rectangle();
        c.draw();
        r.draw();
        System.out.println();
        System.out.println("------ INTERFACE ------");
        Report rp = new Report();
        rp.print();
    }
}
class Book{
    private int bookId;
    private String bookName;
    private String author;
```

```
private double price;
// Default Constructor
public Book(){
    bookId = 0;
    bookName = "";
    author = "";
    price = 0;
}
// Parameterized Constructor
public Book(int id, String name, String author, double price){
    this.bookId = id;
    this.bookName = name;
    this.author = author;
    this.price = price;
}
// Getters
public int getBookId(){
    return bookId;
}
public String getBookName(){
    return bookName;
}
public String getAuthor(){
    return author;
}
public double getPrice(){
    return price;
```

```
    }
    // Setters
    public void setBookId(int id){
        bookId = id;
    }
    public void setBookName(String name){
        bookName = name;
    }
    public void setAuthor(String a){
        author = a;
    }
    public void setPrice(double p){
        price = p;
    }
    public void display(){
        System.out.println("Book ID : " + bookId);
        System.out.println("Book Name : " + bookName);
        System.out.println("Author : " + author);
        System.out.println("Price : " + price);
    }
}
class Person{
    String name;
    int age;
    public Person(String name,int age){
        this.name=name;
        this.age=age;
```

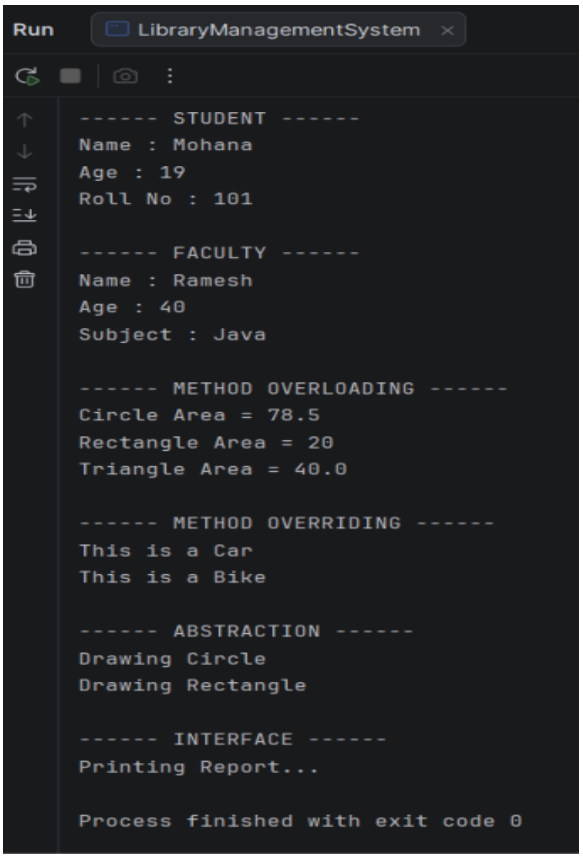
```
    }
    public void display(){
        System.out.println("Name : "+name);
        System.out.println("Age : "+age);
    }
}
class Student extends Person{
    int rollNo;
    public Student(String name,int age,int rollNo){
        super(name,age);
        this.rollNo=rollNo;
    }
    public void display(){
        super.display();
        System.out.println("Roll No : "+rollNo);
    }
}
class Faculty extends Person{
    String subject;
    public Faculty(String name,int age,String subject){
        super(name,age);
        this.subject=subject;
    }

    public void display(){
        super.display();
        System.out.println("Subject : "+subject);
```

```
    }
}
class Area{
    public double calculateArea(double radius){
        return 3.14*radius*radius;
    }
    public int calculateArea(int length,int breadth){
        return length*breadth;
    }
    public double calculateArea(double base,double height){
        return 0.5*base*height;
    }
}
class Vehicle{
    public void display(){
        System.out.println("This is a Vehicle");
    }
}
class Car extends Vehicle {
    @Override
    public void display() {
        System.out.println("This is a Car");
    }
}
class Bike extends Vehicle{
    @Override
    public void display(){
```

```
        System.out.println("This is a Bike");
    }
}
abstract class Shape{
    abstract void draw();
}
class Circle extends Shape{
    @Override
    void draw(){
        System.out.println("Drawing Circle");
    }
}
class Rectangle extends Shape{
    @Override
    void draw(){
        System.out.println("Drawing Rectangle");
    }
}
interface Printable{
    void print();
}
class Report implements Printable{
    @Override
    public void print(){
        System.out.println("Printing Report...");
    }
}
```

**Output :**


Run
LibraryManagementSystem
------ STUDENT ------
Name : Mohana
Age : 19
Roll No : 101
------ FACULTY ------
Name : Ramesh
Age : 40
Subject : Java
------ METHOD OVERLOADING ------
Circle Area = 78.5
Rectangle Area = 20
Triangle Area = 40.0
------ METHOD OVERRIDING ------
This is a Car
This is a Bike
------ ABSTRACTION ------
Drawing Circle
Drawing Rectangle
------ INTERFACE ------
Printing Report...
Process finished with exit code 0